Light

GCSE Computer Science – Data Representation: Media (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

How many different characters can 7-bit ASCII represent?

- **A.** 64
- **B.** 128
- **C.** 256
- **D.** 512

Q2 (1 mark)

In a bitmap image, what does each pixel store?

- **A.** A colour value

- ○ **B.** A letter or number
- ○ **C.** A sound sample
- ○ **D.** A mathematical formula

Q3 (1 mark)

What process converts analogue sound into digital data?

- ○ **A.** Amplification
- ○ **B.** Printing
- ○ **C.** Broadcasting
- ○ **D.** Sampling

Q4 (1 mark)

What is a key feature of Unicode?

- ○ **A.** It only works with English text
- ○ **B.** It can represent characters from virtually all world languages
- ○ **C.** It uses fewer bits than ASCII
- ○ **D.** It is only used for numbers

Q5 (1 mark)

What does the resolution of a bitmap image describe?

- ○ **A.** The number of pixels in the image (width x height)
- ○ **B.** The file format used
- ○ **C.** The brightness of the image
- ○ **D.** The age of the image

Q6 (1 mark)

What unit is used to measure sampling rate?

- ○ **A.** Bytes
- ○ **B.** Pixels
- ○ **C.** Hertz (Hz)
- ○ **D.** Metres

Q7 (1 mark)

An image with 1-bit colour depth can display how many colours?

- ○ **A.** 1 colour
- ○ **B.** 256 colours
- ○ **C.** 8 colours
- ○ **D.** 2 colours

Q8 (1 mark)

In ASCII, the code for 'A' is 65. What type of characters have codes from 48 to 57?

- ○ **A.** Lowercase letters
- ○ **B.** Uppercase letters
- ○ **C.** Digit characters (0-9)
- ○ **D.** Punctuation marks

Q9 (1 mark)

What does increasing the sample resolution (bit depth) of an audio recording improve?

- ○ **A.** The accuracy of each amplitude measurement
- ○ **B.** The length of the recording
- ○ **C.** The number of audio channels
- ○ **D.** The speed of playback

Q10 (1 mark)

What two pieces of information are needed to calculate the file size of a bitmap image?

- ○ **A.** File name and date created
- ○ **B.** Camera model and lens type
- ○ **C.** Image brightness and contrast

- ○ **D.** Number of pixels and colour depth

Q11 (1 mark)

For character codes 0 to 127, what is the relationship between ASCII and Unicode?

- ○ **A.** They use completely different codes
- ○ **B.** Unicode codes are double the ASCII codes
- ○ **C.** They use identical codes
- ○ **D.** ASCII codes are higher than Unicode codes

Q12 (1 mark)

Why must sound be converted from analogue to digital for computer storage?

- ○ **A.** Digital sound is louder
- ○ **B.** Digital microphones are cheaper
- ○ **C.** Analogue sound is too quiet
- ○ **D.** Computers can only process binary data (0s and 1s)

Q13 (2 marks)

Calculate the file size in bits for a bitmap image that is 200 x 150 pixels with a colour depth of 4 bits.

- ○ **A.** 30,000 bits

- B. 60,000 bits
- C. 120,000 bits
- D. 240,000 bits

Q14 (2 marks)

Calculate the file size in bytes for a 5-second audio clip sampled at 8,000 Hz with 8 bits per sample.

- A. 40,000 bytes
- B. 8,000 bytes
- C. 64,000 bytes
- D. 320,000 bytes

Q15 (2 marks)

In ASCII, if the code for 'M' is 77, what is the code for 'P'?

- A. 78
- B. 79
- C. 80
- D. 81

Q16 (2 marks)

How many bits of colour depth are needed to represent exactly 256 different colours?

- ○ **A.** 4 bits
- ○ **B.** 6 bits
- ○ **C.** 8 bits
- ○ **D.** 16 bits

Q17 (2 marks)

What happens to image quality and file size when the number of pixels is increased?

- ○ **A.** Quality decreases, file size decreases
- ○ **B.** Quality increases, file size decreases
- ○ **C.** Quality decreases, file size increases
- ○ **D.** Quality increases, file size increases

Q18 (2 marks)

In a 1-bit image where 0 = white and 1 = black, the binary 1111 represents which row of pixels?

- ○ **A.** Four black pixels
- ○ **B.** Four white pixels
- ○ **C.** Two white and two black pixels

- ○ **D.** One black and three white pixels

Q19 (2 marks)

What effect does increasing the sampling rate have on digital audio?

- ○ **A.** The audio becomes quieter
- ○ **B.** The audio plays back faster
- ○ **C.** The audio file becomes smaller
- ○ **D.** The audio captures more detail of the original sound wave

Q20 (2 marks)

Why would a website need to use Unicode instead of ASCII?

- ○ **A.** To make the website load faster
- ○ **B.** To display content in languages like Chinese, Arabic, or Japanese
- ○ **C.** To reduce storage costs
- ○ **D.** To prevent the website from crashing

Q21 (2 marks)

A graphic designer changes an image from 24-bit colour to 8-bit colour. What happens?

- ○ **A.** The image has more colours and a larger file
- ○ **B.** The image file format changes automatically
- ○ **C.** The image dimensions change
- ○ **D.** The image has fewer colours and a smaller file

Q22 (2 marks)

A camera takes photos at 4000 x 3000 pixels. How many megapixels is this?

- ○ **A.** 7 megapixels
- ○ **B.** 12 megapixels
- ○ **C.** 14 megapixels
- ○ **D.** 24 megapixels

Q23 (2 marks)

In ASCII, '0' (the digit character) has code 48. What is the code for the character '7'?

- ○ **A.** 7
- ○ **B.** 55
- ○ **C.** 57
- ○ **D.** 71

Q24 (3 marks)

A digital photo is 1200 x 800 pixels with 24-bit colour depth. Calculate the file size in megabytes (MB).

- ○ **A.** 2.88 MB
- ○ **B.** 23.04 MB
- ○ **C.** 28.8 MB
- ○ **D.** 230.4 MB

Q25 (3 marks)

A voice recorder captures audio at 11,025 Hz sampling rate with 8-bit resolution for 2 minutes. Calculate the file size in kilobytes (kB).

- ○ **A.** 1,323 kB
- ○ **B.** 10,584 kB
- ○ **C.** 1,058.4 kB
- ○ **D.** 132.3 kB

Q26 (3 marks)

Draw the 4x4 pixel image represented by the 1-bit binary data: 1001 0110 0110 1001 (where 1 = black, 0 = white). Describe the pattern.

- ○ **A.** A vertical line down the middle
- ○ **B.** A horizontal line across the middle
- ○ **C.** An X shape crossing the image diagonally

- ○ **D.** A square border around the edge

Q27 (3 marks)

An image editor shows a 500 x 400 pixel image with file size of 200 kB. Calculate the colour depth being used.

- ○ **A.** 4 bits
- ○ **B.** 8 bits
- ○ **C.** 16 bits
- ○ **D.** 24 bits

Q28 (3 marks)

Recording A uses 22,050 Hz and 8-bit resolution. Recording B uses 44,100 Hz and 16-bit resolution. Both are 1 minute long. How many times larger is Recording B?

- ○ **A.** 2 times larger
- ○ **B.** 4 times larger
- ○ **C.** 6 times larger
- ○ **D.** 8 times larger

Q29 (3 marks)

A text file contains the message 'HELLO WORLD' (11 characters including the space). Using 7-bit ASCII, calculate the file size in bits, then explain why the actual stored size might be different.

- ○ **A.** 77 bits; stored size may be larger because systems often use 8 bits per character for alignment
- ○ **B.** 88 bits; stored size is always exactly 88 bits
- ○ **C.** 11 bits; stored size may be larger due to file headers
- ○ **D.** 110 bits; stored size may be smaller due to automatic shortening

Q30 (3 marks)

A game needs sprite images that are exactly 64 x 64 pixels. The maximum memory for each sprite is 2 kB. What is the maximum colour depth that can be used?

- ○ **A.** 2 bits (4 colours)
- ○ **B.** 3 bits (8 colours)
- ○ **C.** 4 bits (16 colours)
- ○ **D.** 8 bits (256 colours)

Total: **55** marks • Time: **30 min**

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